PRODUCTION AI CASE STUDY

ClinicDocs AI

HEALTHCARE CONVERSATIONAL RAG

Why 90% of RAG bots fail in production — and how I built one that doesn't.

Most tutorials crash under tight server memory, leak patient files across sessions, and hallucinate under trick queries. Here is how I engineered past every bottleneck.

clinicdocs-ai.streamlit.app

✓ 10/10 EVALS PASSED

PATIENT / USER QUERY:

"What are the clinic operating hours on Friday?"

CLINICDOCS AI (VERIFIED GROUNDED):

0.82s Latency

On Friday, Noor Care Clinic operates in two shifts:

- **Morning:** 9:00 AM – 1:00 PM
- **Evening:** 3:00 PM – 8:00 PM

(Clinic is closed between 1:00 PM and 3:00 PM for Friday prayers)

Noor_Care_Clinic_FAQ.pdf Page 1

Similarity: 78.2%

-70%	< 25ms	$0.00
RAM Footprint	**Refusal Gate**	**Off-Topic Cost**
Quantized ONNX Engine	Instant Vector Filter	0 Tokens Wasted

Architected & Built by **Farwa Ali Nasir** • AI & ML Engineer

Swipe for the breakdown →

SYSTEM ENGINEERING

The 3 Bottlenecks

WHY TUTORIALS LIE TO YOU

Tutorial Demos vs Production Systems

Standard tutorial RAG breaks under strict container resource limits, multi-tenant isolation requirements, and off-topic questions. Here is how I solved each:

01 • THE PYTORCH MEMORY BLOAT TRAP

SOLUTION: FastEmbed ONNX

The Trap: Default `sentence-transformers` loads ~800MB PyTorch wheels. In 1GB serverless containers, cold starts crash instantly from Out-Of-Memory (OOM).

✓ **My Production Fix:** Quantized ONNX runtime (`bge-small-en-v1.5`). PyTorch stripped completely — cold starts in **2.22s** and peaks at only **423MB RAM** (70% savings).

02 • MULTI-TENANT DATA LEAKS

SOLUTION: Ephemeral RAM

The Trap: Saving vector collections to local disk files risks leaking confidential clinic documents across sessions and fills container disk space over time.

✓ **My Production Fix:** Pure in-memory vector store (`chromadb.EphemeralClient`) isolated by session UUIDs with automated LRU memory eviction. Zero disk traces left behind.

03 • HALLUCINATIONS & WASTED API SPEND

SOLUTION: Vector Math Gate

The Trap: Passing off-topic questions directly to LLMs burns costly API tokens and risks generating dangerous medical hallucinations.

✓ **My Production Fix:** Cosine similarity threshold (`s >= 0.48`). Unrelated queries are rejected in **18ms** before invoking the LLM — **$0.00 billed**.

DETERMINISTIC ARCHITECTURE

Engineered for Sub-Second Speed and Zero Data Leaks

Every single pipeline step uses strict type validation, custom exception hierarchies, and automated Pydantic schema validation:

STAGE 01 — src/ingest.py

Deterministic Ingest

- **1-Indexed Offset:** PyPDF 0-index corrected to physical document page numbers.
- **SHA-256 Hashing:** Byte checksums skip redundant document re-indexing.
- **Scanned Guard:** Rejects blank scanned image PDFs gracefully.

STAGE 02 — FastEmbed ONNX

Quantized Vectors

- **Model:** BAAI/bge-small-en-v1.5 quantized embeddings (384-dimensions).
- **Zero PyTorch:** Completely eliminates 800MB heavyweight wheels.
- **Cold Start:** 2.22s embedding init on a standard single-core CPU.

STAGE 03 — src/retriever.py

In-Memory Isolation

- **Zero Disk Retention:** Collections live in RAM via EphemeralClient.
- **Session Privacy:** Dedicated collection keys per user UUID.
- **LRU Eviction:** Old sessions purged after 10 active collections.

STAGE 04 — src/chain.py

Two-Layer Guardrails

- **Layer 1 Math Gate:** Cosine similarity halts off-topic queries in 18ms.
- **Layer 2 Grounding:** XML prompt sandbox blocks injection attacks.
- **Smart Rewriting:** Condenses history only from Turn 2 onwards.

VERIFIED BEHAVIOR

Live Query Tests

LIVE INTERROGATION TEST

Strict Grounding vs Zero-Token Refusal

Two real side-by-side test cases showing how the system answers clinical policy questions versus rejecting out-of-scope trick queries:

✓ TEST 1: IN-DOMAIN CLINICAL QUERY

Latency: 0.94s • 100% Grounded

User Query: "How much is a specialist consultation, and what is the follow-up policy?"

A consultation with a Specialist Doctor (Cardiologist, Pediatrician, Dermatologist) costs **PKR 3,000**. Follow-up visits within 7 days of the initial consultation receive a **50% discount (PKR 1,500)**.

Clinic_Fee_Schedule_2026.pdf — Page 2

Cosine Match: 81.4%

TEST 2: OUT-OF-DOMAIN TRICK QUERY

18ms Refusal • $0.00 Spent

User Query: "What is the capital city of France?"

"I couldn't find this in the uploaded documents."

Max observed cosine similarity: **`0.361 < 0.48 threshold`**. The query was intercepted at the vector stage. Zero LLM API calls made. **0 tokens billed.**

MEASURED TELEMETRY

Production Benchmarks

SINGLE CORE CPU TELEMETRY

Hard Data Beats Portfolio Claims

Measured live under container resource constraints using automated telemetry scripts:

ENGINEERING METRIC	CLINICDOCS AI	NAIVE PYTORCH RAG	IMPACT
Embedding Model RAM	136.2 MB	850+ MB	-84% Less RAM
Total Process Peak RAM	423.2 MB	1,420 MB (OOM Risk)	Fits 1GB Cap
Out-of-Domain Refusal	< 25 ms	2,850 ms (Full LLM)	99% Faster
Cost on Trick Questions	$0.00 (0 Tokens)	Standard API Rates	Zero Cost
Benchmark Eval Score	10 / 10 Passed	Hallucinations	100% Grounded

FAN

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Open to Full-Time & Associate AI/ML & SWE Roles

Live Demo & Code 👇

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